

SEMESTRAL DISTRIBUTION WITH COREQUISITES

**BACHELOR OF SCIENCE IN ELECTRICAL ENGINEERING (BSEE)
(LIST OF COURSES BY SEMESTER)
BOR Resolution No. 357, Series of 2017**

First Year, First Semester

Course Code	Course Title	Units	Hrs/Wk			Prerequisite(s)	Corequisite(s)
			Lec	Lab	Total		
MAT060	Calculus with Analytical Geometry 1	4	4	0	4	none	none
CHM012	Chemistry for Engineers	3	3	0	3	none	CHM012.1
CHM012.1	Chemistry for Engineers Laboratory	1	0	3	3	none	CHM012
ENS150.2	Engineering Graphics	2	0	6	6	none	none
GEC104	Mathematics in the Modern World	3	3	0	3	none	none
GEC101	Understanding the Self	3	3	0	3	none	none
GEC105	Readings in Philippine History	3	3	0	3	none	none
NST001	National Service Training Program 1	-3	3	0	3	none	none
PED001	Exercise Prescription and Management	2	2	0	2	none	none
Total		21 (24)	21	9	30		

First Year, Second Semester

Course Code	Course Title	Units	Hrs/Wk			Prerequisite(s)	Corequisite(s)
			Lec	Lab	Total		
MAT070	Calculus with Analytical Geometry 2	4	4	0	4	MAT060	none
PHY052	Physics for Engineers	3	3	0	3	MAT060	MAT070; PHY052.1
PHY052.1	Physics for Engineers Laboratory	1	0	3	3	MAT060	PHY052
EEE100	Basic Programming for Electrical Engineering	1	0	3	3	none	none
GEC102	Purposive Communication	3	3	0	3	none	none
GEC109	Life and Works of Rizal	3	3	0	3	none	none
FIL103	Inobasyon sa Wikang Filipino	3	3	0	3	none	none
NST002	National Service Training Program 2	-3	3	0	3	NST001	none
PED002	Dance/Martial Arts	2	2	0	2	PED001	none
FPE101	Fundamentals of Peace Education	3	3	0	3	none	none
Total		23 (26)	24	6	30		

Second Year, First Semester

Course Code	Course Title	Units	Hrs/Wk			Prerequisite(s)	Corequisite(s)
			Lec	Lab	Total		
ENS181	Engineering Mathematics	3	3	0	3	MAT070	none
EEE130	Electrical Circuit Theory 1	3	3	0	3	MAT070; PHY052; PHY052.1	EEE130.1
EEE130.1	Electrical Circuit Theory 1 Laboratory	1	0	3	3	MAT070; PHY052; PHY052.1	EEE130
ENS161	Statics of Rigid Bodies	3	3	0	3	MAT070, (PHY052; PHY052.1) / (PHY061, PHY061.1)	none
GEC106	Art Appreciation	3	3	0	3	none	none
GEC107	Ethics	3	3	0	3	none	none
MEC121	General Thermodynamics and Heat Transfer	3	3	0	3	PHY052, PHY052.1	None
ENS185	Engineering Data Analysis	3	3	0	3	GEC104, MAT070	none
PED003	Individual/Dual Sports/Traditional/ Recreational Games	2	2	0	2	PED001	none
Total		24	23	3	26		

Second Year, Second Semester

Course Code	Course Title	Units	Hrs/Wk			Prerequisite(s)	Corequisite(s)
			Lec	Lab	Total		
EEE101	Engineering Mathematics for Electrical Engineering	3	3	0	3	ENS181	none
ENS162	Dynamics of Rigid Bodies	3	3	0	3	ENS161	none
ENS164	Mechanics of Deformable Bodies	3	3	0	3	ENS161	none
EEE132	Electrical Circuit Theory 2	3	3	0	3	EEE130, EEE130.1	EEE132.1
EEE132.1	Electrical Circuit Theory 2 Laboratory	1	0	3	3	EEE130, EEE130.1	EEE132
EEE134	Electronic Circuits: Devices and Analysis	3	3	0	3	EEE130, EEE130.1	EEE134.1
EEE134.1	Electronic Circuits: Devices and Analysis Laboratory	1	0	3	3	EEE130, EEE130.1	EEE134
EEE136	Engineering Electromagnetics	3	3	0	3	PHY052, PHY052.1, ENS181	none
EEE192	Electrical Engineering Law, Codes, and Professional Ethics	2	2	0	2	GEC107	none
PED004	Team Sports	2	2	0	2	PED001	none
Total		24	22	6	28		

Third Year, First Semester

Course Code	Course Title	Units	Hrs/Wk			Prerequisite(s)	Corequisite(s)
			Lec	Lab	Total		
EEE102	Numerical Methods and Analysis for Electrical Engineering	3	2	3	5	ENS181	none
ENS167	Fundamentals of Materials Science and Engineering	3	3	0	3	(CHM012, CHM012.1) / (CHM014, CHM014.2)	none
EEE180	Logic Circuits and Switching Theory	3	3	0	3	EEE134, EEE134.1	EEE180.1
EEE180.1	Logic Circuits and Switching Theory Laboratory	1	0	3	3	EEE134, EEE134.1	EEE180
ENS191	Engineering Economy	3	3	0	3	GEC104	ENS185 / BET120
EEE176	Industrial Electronics	3	3	0	3	EEE134, EEE134.1	EEE176.1
EEE176.1	Industrial Electronics Laboratory	1	0	3	3	EEE134, EEE134.1	EEE176
EEE186	Fundamentals of Electronic Communications	3	3	0	3	EEE134, EEE134.1	none
EEE193	Electrical Standards and Practices	1	0	3	3	EEE192	none
EEE146	Electrical Machines 1	2	2	0	2	EEE132, EEE132.1, EEE136	none
Total		23	19	12	31		

Third Year, Second Semester

Course Code	Course Title	Units	Hrs/Wk			Prerequisite(s)	Corequisite(s)
			Lec	Lab	Total		
ENS190	Basic Occupational Safety and Health	3	3	0	3	none	none
EEE184	Microprocessor, Microcontroller Systems and Design	3	3	0	3	EEE180, EEE180.1	EEE184.1
EEE184.1	Microprocessor, Microcontroller Systems and Design Laboratory	1	0	3	3	EEE180, EEE180.1	EEE184
EEE144	Electrical Apparatus and Devices	2	2	0	2	EEE132, EEE132.1	EEE144.1
EEE144.1	Electrical Apparatus and Devices Laboratory	1	0	3	3	EEE132, EEE132.1	EEE144
EEE148	Electrical Machines 2	3	3	0	3	EEE146	EEE148.1
EEE148.1	Electrical Machines 2 Laboratory	1	0	3	3	EEE146	EEE148
HIS003	History of Filipino Muslims and Indigenous Peoples of Mindanao	3	3	0	3	none	none
FIL101	Wika at Kultura sa Mapayapang Lipunan	3	3	0	3	none	none
EEE160	Feedback and Control Systems	3	3	0	3	EEE101, EEE134, EEE134.1	EEE160.1
EEE160.1	Feedback and Control Systems Laboratory	1	0	3	3	EEE101, EEE134, EEE134.1	EEE160
Total		24	20	12	32		

Third Year, Summer Term

Course Code	Course Title	Units	Hours/Term			Prerequisite(s)	Corequisite(s)
			Lec	Lab	Total		
EEE197	On-the-Job Training	3	0	240	240	ENS190	none
Total		3	0	240	240		

Fourth Year, First Semester

Course Code	Course Title	Units	Hrs/Wk			Prerequisite(s)	Corequisite(s)
			Lec	Lab	Total		
ENS165	Mechanics of Fluids	3	3	0	3	ENS162	none
EEE150	Power System Analysis	3	3	0	3	EEE193	EEE150.1
EEE150.1	Power System Analysis Laboratory	1	0	3	3	EEE193	EEE150
EEE152	Distribution Systems and Substation Design	2	2	0	2	none	EEE150; EEE150.1; EEE152.1
EEE152.1	Distribution Systems and Substation Design Laboratory	1	0	3	3	none	EEE150; EEE150.1; EEE152
EEE154	Fundamentals of Power Plant Engineering Design	1	0	3	3	none	EEE150; EEE150.1
Elective 1	(Technical Elective 1)	3	3	0	3	(As the course requires)	(As the course requires)
EEE198	Research Methods	1	0	3	3	ENS185	none
EEE103	Management of Engineering Projects	2	2	0	2	ENS191	none
EEE178	Instrumentation and Control	3	3	0	3	EEE160, EEE160.1	EEE178.1
EEE178.1	Instrumentation and Control Laboratory	1	0	3	3	EEE160, EEE160.1	EEE178
Total		21	16	15	31		

Fourth Year, Second Semester

Course Code	Course Title	Units	Hrs/Wk			Prerequisite(s)	Co-requisite(s)
			Lec	Lab	Total		
GEC103	The Contemporary World	3	3	0	3	none	none
GEC108	Science, Technology and Society	3	3	0	3	none	none
EEE158	Electrical Systems and Illumination Engineering Design	3	3	0	3	EEE148, EEE148.1	EEE158.2
EEE158.2	Electrical Systems and Illumination Engineering Design	2	0	6	6	EEE148, EEE148.1	EEE158
Elective 2	(Technical Elective 2)	3	3	0	3	(As the course requires)	(As the course requires)
EEE199	Undergraduate Thesis	1	0	3	3	EEE198	none
EEE196	Seminars/Colloquia	1	0	3	3	none	none
ENS193	Technopreneurship for Engineers and Technologists	3	3	0	3	ENS191	ENS192
ENS195	Environmental Science and Engineering with Safety	3	3	0	3	(CHM012, CHM012.1) / (CHM013, CHM013.2) / (CHM018, CHM018.1)	none
Total		22	18	12	30		

GRAND TOTAL: 185 (191) Units
Suggested Free or Track Elective Track Subjects

Course Code	Course Title	Units	Hrs/Wk			Prerequisite(s)	Corequisite(s)
			Lec	Laboratory	Total		
A. Power System Protection Track							
EEE127	Protection of Generators,Transformers, Bus-bars and Lines	3	3	0	3	EEE148	none
EEE128	Protective Relaying	3	3	0	3	EEE150	none
EEE129	Surge Protection in Power System	3	3	0	3	EEE150	none
B. Power System Operation Track							
EEE155	Power System Reliability	3	3	0	3	EEE193	none
EEE156	Power System Dynamics and Stability	3	3	0	3	EEE150	none
EEE157	Power System Operation and Control	3	3	0	3	EEE150	none
C. Power System Economics							
EEE151	Power System Market Operation	3	3	0	3	EEE193	none
EEE153	Economic Operation of Power System	3	3	0	3	EEE150	none
EEE159	Power Quality and Demand Side Management (DSM)	3	3	0	3	EEE150	none
D. Advanced Power System Analysis and Design Track							
EEE105	Transmission and Distribution	3	3	0	3	EEE193	none
EEE107	Power Substation	3	3	0	3	EEE193	none

E. Advanced Electrical Systems Design Track							
EEE108	High Rise Building Design	3	3	0	3	EEE193	none
EEE110	Substation Design	3	3	0	3	EEE193	none
EEE112	High-Voltage Underground Cable Design (AC/DC Systems)	3	3	0	3	EEE193	none
F. Entrepreneurship Track							
EEE109	Project Management	3	3	0	3	ENS191	none
EEE111	Project Acceptance, Testing and Documentation	3	3	0	3	ENS191	none
EEE113	Sales and Marketing Management	3	3	0	3	ENS191	none
G. Machine Automation and Process Control Track							
EEE163	Pneumatics & Process Control	3	3	0	3	EEE160	none
EEE164	Electropneumatics	3	3	0	3	EEE160	none
EEE165	Programmable Logic Controllers in Manufacturing	3	3	0	3	EEE160	none
F. Special Studies in Renewable Energy Resources							
EEE115	Solar and Wind Energy	3	3	0	3	EEE148	none
EEE117	Hydro and Waves/Ocean Energy	3	3	0	3	EEE148	none
EEE119	Biomass Energy	3	3	0	3	EEE148	none

Note: The offering of the track electives will be based on the availability of the resources (faculty force) and the number of students who will request for it.